Keahooi Hung

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Backend engineer with 5 years of experience building and operating security-critical authorization infrastructure at scale. Proven track record designing public APIs, leading cross-team initiatives, and owning compliance and disaster recovery efforts for a platform handling tens of millions of requests per day.

SKILLS

Languages: Java, Go, SQL, Python, HCL (Terraform), Bash, Protobuf/gRPC

Infrastructure: AWS (RDS, IAM, KMS, ECR), Kubernetes, Apache Kafka, PostgreSQL, HashiCorp Vault, Docker, Helm

Frameworks: Guice, JOOQ, Jersey, Go-kit, Maven

Tools: Git, GitHub, Datadog, Grafana, OpenSearch, LaunchDarkly, CI/CD (SemaphoreCI)

EXPERIENCE

Senior Software Engineer II, *Confluent* *June 2021 — Present*

- Access Management team, Cloud Platform. Building deep domain expertise in cloud authorization infrastructure.
- Owned and developed core authorization services for Confluent Cloud, the fully managed Apache Kafka platform. These authorization services sit in the critical path of every API call across the platform, handling ~45 million requests/day.
- Mentored an engineer through his internship (L1) and supported his growth to L2 and L3 (Senior Engineer promotion).

Software Engineering Intern, *Confluent* *May 2020 — Aug 2020*

Undergraduate Student Instructor, *UC Berkeley EECS* Jan 2020 — May 2020

PROJECTS

IP Filtering for Kafka (2024-2025)

- Led a cross-team project extending Confluent Cloud's application-layer IP filtering to Apache Kafka clusters, the #1 ranked project on the 2025 company-wide product initiative stack rank.
- Designed and built a CDC (Change Data Capture) pipeline that propagates IP filter policies from the control plane database to thousands of distributed Kafka clusters in real time.
- Identified a critical integration issue early—internal service-to-service traffic would be incorrectly blocked—giving the team enough runway to redesign and meet the customer deadline. Recognized as a Quarterly Key Contributor.

FedRAMP Moderate Compliance (2023-2024)

- Sole engineer responsible for achieving FedRAMP Moderate compliance for the team's four authorization services. This work unlocked the U.S. federal customer market for Confluent Cloud.
- Addressed federal cryptographic requirements, migrated to hardened container images, remediated all CVEs, and authored compliance-specific end-to-end tests.

Business Continuity & Disaster Recovery (2023-2024)

- Sole engineer responsible for building multi-region disaster recovery infrastructure for the team's authorization services, enabling Confluent Cloud's control plane to survive a full AWS region failure.
- Deployed authorization services to a secondary AWS region. Integrated services with a region-aware failover SDK, migrated database authentication to AWS IAM, and refactored Terraform modules for multi-region support.
- Represented the team during company-wide failover drills and authored disaster recovery runbooks.

Control Plane Authorization Service Extraction (2022)

- Sole architect of extracting the critical path cloud RBAC authorization service from a legacy Apache Kafka broker plugin into a standalone, independently deployable Kubernetes microservice for Confluent Cloud. This decoupled our release process from Kafka, reducing the end to end deployment from weeks to days.
- Designed and implemented the standalone service packaging, Kubernetes deployment configuration, DNS routing, load balancer setup, and full observability stack (metrics, logging, dashboards, alerts).
- Orchestrated the live traffic switchover with zero downtime

Role binding CRUD Public API (2022)

- Sole architect and primary implementer of the RBAC Public REST API for Confluent Cloud — the interface used by customers, the Confluent CLI, and internal services to manage access control. The API manages millions of role bindings in production.

EDUCATION

University of California, Berkeley *May 2021*

BS, Electrical Engineering & Computer Sciences. GPA: **3.94,** 14 upper division courses (the degree requirement is 5)

HONORS

Eta Kappa Nu, Tau Beta Pi, Dean's List, Eagle Scout